



Published in final edited form as:

J Orthop Res. 2018 May ; 36(5): 1478–1486. doi:10.1002/jor.23770.

Longitudinal changes in knee gait mechanics between 2 and 8 years after anterior cruciate ligament reconstruction

Jennifer C. Erhart-Hledik, Ph.D.^a, Constance R. Chu, M.D.^{a,c}, Jessica L. Asay, M.S.^{b,c}, and Thomas P. Andriacchi, Ph.D.^{a,b,c}

^aDepartment of Orthopaedic Surgery, Stanford University Medical Center, Stanford, CA

^bDepartment of Mechanical Engineering, Stanford University, Stanford, CA

^cVeterans Affairs Palo Alto Health Care System, Palo Alto, CA

Abstract

The purpose of this study was to longitudinally investigate changes in knee joint kinematics and kinetics from 2 to 8 years post-ACLR. Seventeen subjects with primary unilateral transtibial ACLR performed bilateral gait analysis approximately 2 years and 8 years post-ACLR. Seventeen matched healthy control subjects were also analyzed. Kinematic and kinetic comparisons between the ACLR and contralateral limbs over time were completed using a 2x2 (time, limb) repeated-measures ANOVA. Unpaired Student's t-tests were used to compare the ACLR and contralateral kinematics and kinetics to the control group. The ACLR and contralateral limbs had similar gait changes over time. Kinetic changes over time included a reduction in first ($p=0.048$) and second ($p<0.001$) peak extension moments, internal rotation moment ($p<0.001$), adduction moment (first peak: $p=0.002$, second peak: $p=0.009$, impulse: $p=0.004$) and an increase in peak knee flexion moment ($p=0.002$). Kinematic changes over time included increases in peak knee flexion angle in the first half of stance ($p=0.026$), minimum knee flexion angle in the second half of stance ($p<0.001$), and average external rotation angle during stance ($p=0.007$), and a reduction in average anterior femoral displacement during stance ($p=0.006$). Comparison to healthy controls demonstrated improvement in some gait metrics over time. The results demonstrated longitudinal changes from 2 to 8 years after ACLR in knee joint kinetics and kinematics that have been related to clinical outcome after ACLR and the progression of knee OA, and support future larger and comprehensive investigations into long-term changes in joint mechanics in the ACLR population.

Keywords

ACL; Reconstruction; Gait Analysis; Knee; Kinematics; Kinetics

Correspondence To: Dr. Jennifer Erhart-Hledik, Department of Orthopaedic Surgery, Stanford University, Stanford, CA 94305-4308, Tel.: +1 650-723-5793, Fax: +1 650-725-1587, jerhart@stanford.edu.

Author Contributions Statement: Dr. Erhart-Hledik participated in conceptualization of the research study, acquisition, analysis, and interpretation of the data, and drafting of the manuscript. Drs. Andriacchi and Chu each participated in conceptualization of the research study, interpretation of the data, and drafting of the manuscript. Ms. Asay participated in acquisition and analysis of the data and drafting of the manuscript. Dr. Erhart-Hledik takes responsibility for content of the manuscript. All authors have read and approved the final submitted manuscript.

INTRODUCTION

Anterior cruciate ligament (ACL) injury is most often treated with surgical reconstruction, aiming to restore knee anatomy, joint stability, and re-establish normal knee function. Despite the success of ACL reconstruction (ACLR) in returning the majority of patients to normal knee function¹, ACLR does not reduce the risk of development of premature knee osteoarthritis (OA) after ACL injury. Studies report between 39% and 56% of young individuals 30 years of age or younger at the time of trauma develop radiographic knee OA approximately 8–12 years after ACLR surgery^{2–4}. Further, nearly 70% show cartilage degeneration on MRI less than 3 years after ACLR⁵. A proposed mechanism for increased risk of development of knee OA after ACL injury and reconstruction is the evidence of abnormal knee biomechanics beginning early after ACLR as summarized in several recent systematic reviews^{6; 7}. Data demonstrates that cartilage becomes conditioned to a typical loading environment⁸. Therefore, changes in knee kinematics and kinetics during repetitive tasks such as walking have the potential to change the loading conditions within the cartilage and lead to poorer clinical outcomes, cartilage degeneration, and development of knee OA after ACLR^{9–14}.

While kinematic^{7; 15; 16} and kinetic^{6; 7; 17} changes to the ACLR knee have been described in all three planes of motion, the majority of studies have investigated joint biomechanics at a single time point following ACLR. Measurements at a single time point may not capture longer-term outcome of ACLR surgery, including changes that could occur in joint loading over time. For example, conflicting reports regarding changes in the knee adduction moment (KAM) after ACLR have been reported, with studies finding increases¹⁸, no change¹⁹, and decreases^{17; 20; 21} in KAM as compared to the contralateral limb and healthy controls. Such conflicting reports may be due, in part, to changes in joint biomechanics with time after ACLR.

Despite the potential importance of gait changes after ACLR, few studies have prospectively looked at changes in gait over time after reconstruction surgery^{22–27}, with the majority of studies investigating changes at a relatively early time point after surgery. To date, the longest prospective analyses have been conducted up to 3 years^{23; 25}, 4 years²², and 5 years post-ACLR²⁴, and have reported conflicting results. Webster et al.²³ investigated three-dimensional kinematic and kinetic changes during walking over a period of 10 months to 3 years after ACLR, and found changes in several gait variables including knee flexion at terminal stance, KAM, and knee extension moment over time. Titchenal et al.²² investigated changes in knee kinematics over a period of 2 to 4 years after ACLR and found changes in the knee center of rotation. Tagesson et al.²⁴ found no changes in dynamic tibial translation during gait over a 5 year time period after ACLR, while Lin et al.²⁵ investigated kinematic changes over a period of 6 months to 3 years post-ACLR during a step-up motion and found longitudinal changes in tibiofemoral contact locations. Given the relative lack of longer-term data on longitudinal three-dimensional kinematic and kinetic changes after ACLR during level walking, a common task that all individuals participate in, and given that the time scale between ACL injury and the development of radiographic knee OA is often greater than 10 years², there is a need for longer term follow-up to understand changes in three-dimensional gait mechanics during walking after ACLR.

Thus, the purpose of this study was to prospectively investigate changes in knee gait mechanics during walking between two and eight years after ACLR. As prior work demonstrates bilateral neuromuscular alterations after ACL injury^{28–31}, longitudinal changes in both the ACLR and contralateral limbs were investigated. Specifically, we hypothesized that there would be longitudinal changes in specific kinematic (knee flexion, adduction, rotation, and femoral translation) and kinetic (adduction, flexion, extension, and rotation moments) features in the ACLR and contralateral limbs during level walking from 2 to 8 years post-ACLR. Data from an age-, sex-, and BMI-matched healthy control group were analyzed for comparison to the ACL cohort.

METHODS

Study Design and Level of Evidence

Prognostic Study, Level II

Participants

This study was approved by the Stanford University Institutional Review Board, and all patients provided written informed consent prior to participation. Seventeen subjects 8 years after unilateral primary transtibial ACLR (7.7 ± 0.7 years) were recruited from a cohort of 42 subjects who performed bilateral gait analysis approximately 2 years post-ACLR (2.2 ± 0.3 years) and agreed to future contact¹⁷. Because the 42 subjects from the cross-sectional study were not informed or consented for the follow-up test, new IRB approval was sought to re-contact these subjects for additional testing. As such, the study population represents an opportunity sample³² consisting of subjects without additional knee injuries or surgeries that were available to return for the 8-year follow-up. Of the subjects who did not return for follow-up gait testing, 9 did not respond, 5 were unable to participate due to time constraints, 3 had moved away from the area, 3 had additional arthroscopic surgery on their affected limb, 2 had re-torn their ACL, 2 were not interested, and 1 had injured their contralateral limb.

Inclusion criteria for the 2-year study¹⁷ included: 1) successful single-bundle unilateral ACLR based on clinical exam (KT-1000 side-to-side difference < 5 mm), 2) no other history of serious lower limb injury, 3) self-reported history of knee stability, and 4) knee MRI to confirm intact graft. Exclusion criteria included: 1) removal of more than 25% of the meniscus, 2) a history of other serious ligamentous injury to either lower limb, 3) clinical instability of the reconstructed knee, 4) BMI > 30 kg/m², 5) significant observable chondral defects by MRI or 6) a history of surgical procedures performed on either lower limb or revision operation of the ACL. The additional inclusion and exclusion criteria for participation in the 8-year follow-up study were willingness to attend an in-person follow-up test and no additional serious injuries or surgical procedures to either lower limb. The ACLR procedures were performed by 7 surgeons, and no participant underwent any further surgery following the reconstruction.

Data from a cohort of 17 healthy control subjects with no history of serious injury to either limb that were age-, sex-, and BMI-matched to the 8-year follow-up cohort were also

analyzed, with one knee randomly selected per control subject. The control cohort was age-matched to the age of the study cohort at 8-year follow-up, as data shows gait is stable before the age of 45 years.³³ The healthy participants provided IRB-approved consent prior to participation.

Gait Analysis

All subjects performed three successful trials of level walking at a self-selected normal speed separately for each limb along a 10-m walkway in their own walking shoes in the same laboratory at the 2 and 8 year time points. Subjects were instructed to walk at their normal walking pace, and variations in walking speed between the three trials were limited by ensuring that the subjects started walking at the same location for the three trials and made the same number of footsteps until they hit the force plate. The order of testing of the limbs was randomly selected. A trial was considered successful if the foot of the leg being tested fully struck the force plate. Kinematic and kinetic data were collected with an optoelectronic motion capture system (Qualisys Medical, Gothenburg, SE) and a multicomponent force plate embedded in the walkway (Bertec Corporation, Columbus, OH), which were synchronized and recorded data at 120 Hz. Lower limb kinematics and kinetics were analyzed with the software application BioMove (Stanford University, CA) using the point cluster technique³⁴ and previously described methods^{34–36}. Briefly, the anatomical frames of the foot, shank, and thigh segments were determined during a standing reference pose collected before the walking trials³⁶. The position and orientation of the foot, shank, and thigh segments were calculated using clusters of reflective markers fixed to the participant. The knee flexion angle, varus rotation angle, and external rotation of the tibia relative to the femur were calculated according to the joint coordinate system³⁷. Anterior displacement of the femur relative to the tibia was defined as the position of the center of the femoral trans-epicondylar axis along the anterior-posterior axis of the tibial anatomical frame. Knee joint moments in all three planes were calculated using inverse dynamics^{38; 39} and were expressed as external moments relative to the tibial anatomical frame based upon the position of palpable anatomical landmarks³⁶. A vertical ground reaction force threshold of 10N was used to detect the beginning and the end of the stance phase (heel-strike and toe-off), and the stance phase for knee kinematics and kinetics were normalized to 101 time points (0% to 100%) during this phase and averaged over the 3 walking trials. Moments were normalized to body weight and height (%Bw*Ht) for comparison between subjects.

Kinematic and kinetic variables for analysis were chosen to provide a comprehensive three-dimensional analysis of changes in gait biomechanics over time, and were based on literature regarding changes in kinematics and kinetics after ACLR^{7; 16; 17; 27; 40; 41}, as well as the importance to clinical outcomes (patient-reported outcomes)^{11; 42; 43} and progression of knee OA^{44–47}. For kinetic analyses, our outcome variables included the peak knee flexion moment (KFM), knee adduction moment first peak (KAM1), second peak (KAM2), and impulse (KAM Impulse), first and second peak extension moments (KEM1 and KEM2), and peak internal (KIRM) and external (KERM) rotation moments. For kinematic analyses, our outcome variables included the peak knee flexion angle during the first half of stance phase, minimum knee flexion angle in the second half of stance phase, peak varus angle in the first half of stance phase, peak valgus angle in the second half of stance phase, average external

rotation angle during stance (heel-strike to toe-off), and average anterior femoral displacement during stance (heel-strike to toe-off). Spatiotemporal variables included walking speed and step length.

Statistical Analysis

Comparisons between the ACLR and contralateral limbs over time were completed using a 2 (limb: ACLR and contralateral) x 2 (time: 2 years post-ACLR and 8 years post-ACLR) repeated-measures analysis of variance (ANOVA) with time and limb as within-subject factors. Upon a significant result of the ANOVA, post hoc paired Student's t-tests were used for analyses. Unpaired Student's t-tests were used to compare the ACLR and contralateral data at baseline and follow-up to the control group. An $\alpha=0.05$ was used for all analyses with the Holm–Bonferroni method to account for multiple comparisons. Adjusted p-values from Holm-Bonferroni analyses are presented throughout the results. All statistical analyses were performed using SPSS version 23.0 (SPSS Inc., Chicago, IL).

RESULTS

Participants

The ACLR study participants included 12 females and 5 males (Table 1). The reconstructions were performed on 9 right knees and 8 left knees. The average ($\pm$ standard deviation) time between injury and reconstruction was 2.0 ± 1.8 months. All ACLR subjects had a single-bundle ACLR performed with a transtibial technique, with 15 performed with an Achilles allograft, one performed with a bone-patellar tendon-bone allograft, and one performed with a bone-patellar tendon-bone autograft. There were no significant changes in patient-reported outcome measures (Lysholm: $p=0.48$; IKDC (International Knee Documentation Committee): $p=0.55$; or KOOS (Knee Injury and Osteoarthritis Outcome Score) Pain: $p=0.71$, Symptoms: $p=0.93$, ADL: $p=0.14$, Sports/Rec: $p=0.38$, QOL: $p=0.35$) between baseline and follow-up as assessed by paired Student's t-tests (Table 1). The control cohort consisted of 12 females and 5 males, with an average ($\pm$ standard deviation) age of 35.4 ± 7.4 years and BMI of 24.5 ± 3.0 kg/m².

Kinetics

There was a significant main effect of time on joint kinetics (Table 2). Over the follow-up period, there was a reduction in the magnitude of KEM1 ($p=0.048$) and KEM2 ($p<0.001$), while an increase in peak KFM ($p=0.002$) was seen (Figure 1). KAM1 ($p<0.001$), KAM2 ($p=0.009$), KAM Impulse ($p=0.004$), and KIRM ($p<0.001$) decreased over follow-up (Figure 1). For all moments, the ACLR and contralateral limb changed similarly over time (no significant interaction effects were observed). No significant effect of limb was seen after controlling for multiple comparisons (Table 2).

Comparison to the healthy control population at baseline demonstrated that the KEM1 was significantly higher in the ACLR ($p=0.012$) and contralateral ($p=0.047$) limbs versus control. Further, at baseline the KERM was significantly reduced in the ACLR ($p=0.041$) limb versus control and the peak KFM was significantly reduced in both the ACLR ($p=0.006$) and

contralateral ($p=0.048$) limbs versus control. At follow-up, KEM2 was significantly reduced in the ACLR ($p=0.011$) limb versus control.

Kinematics

There was a significant main effect of time on joint kinematics (Table 3). For all variables the ACLR and contralateral limb changed similarly over time (no significant interaction effects were observed). Over the follow-up period, there was a significant increase in peak knee flexion angle in the first half of stance phase ($p=0.026$) and in the minimum knee flexion angle in the second half of stance phase ($p<0.001$) (Figure 2). The average external rotation angle increased from 2 to 8 years post-ACLR ($p=0.007$), while a reduction in average anterior femoral displacement (increase in average anterior tibial displacement) was observed ($p=0.006$) (Figure 2). Only one significant effect of limb was seen after controlling for multiple comparisons, with a more flexed ACLR limb versus contralateral limb at the time of minimum knee flexion angle in the second half of stance ($p=0.045$), with a trend toward a more externally rotated tibia over stance on the ACLR limb versus the contralateral limb ($p=0.16$). Comparison to the healthy control population demonstrated only one significant difference, with a more flexed ACLR limb at 8-year follow-up versus control at the time of the minimum knee flexion angle in the second half of stance phase ($p=0.021$).

There were no differences in walking speed or step length between either assessments or legs in the ACLR cohort (Table 3). There was only one significant difference in spatiotemporal variables as compared to the healthy control population, with smaller step length in the contralateral limb at baseline as compared to healthy control ($p=0.04$).

DISCUSSION

The results of this study demonstrate that there are longitudinal changes from 2 to 8 years post-ACLR in knee joint kinetic and kinematic features that have been related to clinical patient-reported outcomes after ACLR^{11; 42; 43} and the progression^{44–47} of knee OA. The ACLR and contralateral limbs changed similarly over the follow-up period, suggesting a bilateral adaptation to injury of a single limb. While it is well-documented that there are alterations in gait mechanics following ACLR, there exists some conflicting evidence on the direction and magnitude of such changes^{6; 7}. These discrepancies may partially be due to the time at which the studies were conducted after reconstruction. Given the long term perspective² in the development of OA in the ACL population, this work is a further step in understanding changes that continue to occur in gait mechanics with time after ACLR.

The results of this study demonstrated an increase in peak KFM from 2 to 8 years post-ACLR. This change could be considered an improvement, as no difference was observed at 8-year follow-up in peak KFM as compared to the healthy control population, while a reduction in peak KFM was observed in both the ACLR and contralateral limbs at baseline versus control. Previous work has shown that ACLR subjects display reduced peak KFM⁶, an indication of diminished quadriceps function⁴⁸, possibly due to reduced quadriceps strength, impaired neuromuscular control, compensatory movements, or lingering symptoms. Thus, the increase in KFM over time may indicate long-term continued recovery of quadriceps function. Further, the reduction in KEM1 and KERM over the follow-up

period may also indicate improvements in joint loading over time, as significant differences were noted at baseline in KEM1 in both the ACLR and contralateral limbs as compared to healthy control and in the KERM in the ACLR limb as compared to healthy control, while no differences were observed at the 8-year follow-up. However, it is unknown if reductions would continue to occur with increasing time post-ACLR (greater than 8-years post-ACLR), and if such changes could be associated with degenerative changes as growing evidence suggests both joint unloading and overloading may be associated with degenerative cartilage changes^{13; 49}. Further, research from early knee OA populations demonstrates reduced joint moments⁵⁰, and a recent study by Wellsandt et al.¹⁰ demonstrated patients developing radiographic knee OA after ACLR had lower KAMs early after reconstruction than patients who did not develop OA. Thus, while no significant differences were observed in KAM1, KAM2, KAM Impulse, or KIRM between ACLR subjects and healthy controls, the reductions in these metrics over time may be indicative of neuromuscular adaptations in loading that precede development of clinical symptoms^{10; 50}. The significant reduction in KEM2 in the ACLR limb over the follow-up period resulted in joint loading that was significantly different from healthy control at follow-up. Prior work has shown a significant reduction in KEM2 in the ACLR limb as compared to the contralateral limb¹⁷, though the significance of this change with regard to development of knee OA is unclear.

No significant differences were noted between the ACLR and contralateral limbs in joint kinetics after controlling for multiple comparisons. Interestingly, the kinetic changes over time in the contralateral limb tended to mimic the pattern of the ACLR limb, with no differences observed in changes over time between the reconstructed and contralateral limbs. The symmetric adaptation over time suggests, consistent with previous work reporting bilateral neuromuscular alterations after ACL injury^{28–31}, that the ACLR patients adapted their gait symmetrically.

Similarly, kinematic changes over time were observed in both the ACLR and contralateral limbs. Analyses demonstrated significant changes (knee flexion-extension, external rotation, and femoral displacement) over time in both the ACLR and contralateral limbs in variables associated with clinical outcomes as assessed by patient-reported outcomes after ACLR^{11; 42; 43} and increased risk of progression⁴⁵ of knee OA. The increase in peak knee flexion over time is consistent with the observed increase in KFM over time, and may suggest an improvement in gait mechanics, as prior work demonstrates that ACLR subjects display reduced midstance peak knee flexion angles during gait⁷. The increase in minimum knee flexion angle in the second half of stance resulted in a significant difference between the ACLR limb and healthy controls at 8-year follow-up, and thus may be indicative of worsening of this kinematic variable. Prior work by Shelbourne and Gray¹⁴ demonstrated that loss of knee extension adversely affected subjective and objective results at a mean of 14 years after surgery, though the clinical significance of this change to the development of knee OA in this population is unclear. The external rotation of the tibia increased over time in both limbs, with a trend towards a more externally rotated tibia in the ACLR limb as compared to the contralateral limb. Prior work in the ACLR population at single time points also shows an increase in external tibial rotation⁴¹. A reduction in anterior femoral displacement was also found, which is consistent with prior work showing an increase in anterior tibial translation (i.e. reduction in anterior femoral displacement) in the ACLR limb

at a single time point^{16; 40}. While no significant differences were observed in tibial rotation or femoral displacement as compared to the healthy control population, the significant changes over time in these variables may suggest worsening of these gait features over the follow-up period, as there is growing evidence that such kinematic changes causing even a small change in load-bearing position can alter loading in the cartilage and lead to degenerative changes in local regions of cartilage not adapted to withstand the new mechanical loading environment^{8; 13; 49}.

It is important to note that while this study demonstrates similar longitudinal kinematic and kinetic changes in the contralateral limb and affected limb after ACLR, the occurrence of premature knee OA in the contralateral knee is much less than in the affected limb⁵¹. This observation suggests that other factors such as altered biology^{52; 53}, in addition to altered biomechanics, play a significant role in development of premature knee OA in the ACLR limb. It is also of interest to note that patient-reported outcome measures (Table 1) showed minimal changes from 2 to 8 years prospectively while gait demonstrated significant changes. This finding suggests that gait may be a more sensitive metric than patient-reported outcomes in patients with well-functioning ACLRs.

The difference in time frames between our study (from 2 to 8 years post-ACLR) and that of other prospective studies of ACLR patients such as Webster et al.²³ and Tagesson et al.²⁴ may explain the different results observed. While we found a reduction in extension in terminal stance and a reduction in extension and adduction moments, Webster et al.²³ found an increase in terminal extension and extension moment in the ACLR limb and an increase in KAM in both limbs from 10 months to 3 years post-ACLR. Further, our work in the current study demonstrated a reduction in anterior femoral displacement over time, while Tagesson et al.²⁴ found no change in maximal anterior tibial translation from 5 weeks to 5 years post-ACLR. The prospective study by Lin et al.²⁵ investigated tibiofemoral cartilage contact locations during a dynamic step-up motion, and found the contact locations were located more anteriorly at 3 years post-ACLR as compared to 6 months post-ACLR, consistent with our finding of a reduction in anterior femoral displacement (an increase in anterior tibial translation) over time. Further work is needed to understand if there is a time at which kinematics and kinetics stabilize after injury and reconstruction and to understand if such changes are related to the development of knee OA. The finding of continued prospective changes in joint mechanics over a long-term period after ACLR suggests that perhaps longer-term rehabilitation or interventional programs may be needed after the 6–12 month period following ACLR during which standard rehabilitation occurs.

There are limitations to the present study which should be acknowledged. The follow-up testing at 8 years was not part of the original study protocol and the number of subjects available at the 8 year follow-up was relatively small. As such the results could be exposed to bias common to opportunity samples³². However, capturing a longer term follow-up in this population shows key mid-term findings of significant longitudinal changes in both kinematic and kinetic measures from 2 to 8 years post-reconstruction that have been related to clinical patient-reported outcomes^{11; 42; 43} after ACLR and knee OA progression^{44–47}. Further, comparison of baseline demographics between subjects who returned for 8 year follow-up and those who did not showed no differences between groups. Because clinical

changes related to knee OA often are not apparent for 10 years or more, these gait changes may show potential in providing earlier warning of deteriorating knee function and opportunities for early intervention. Future work is needed to confirm the results of this study in a larger prospective cohort. While data was collected in the same laboratory with the same set-up at the 2 and 8 year time points, and researchers collecting data at the 2 and 8 year time points were trained by the same single individual with regard to gait analysis techniques, including placement of body markers, there is potential for error associated with gait analysis techniques. Literature⁵⁴ in young healthy adults reported that interrater-intersession standard error of measurement (SEM) values for knee kinematics are typically below 2° with minimally detectable change (MDC) values of approximately 5°. Thus, the significant changes found in the kinematics in present study were above measurement error but generally not large enough to exceed the previously-reported MDC. For knee kinetics, previously-reported SEM values were typically below 0.05 Nm/kg, with MDC values of approximately 0.1 Nm/kg.⁵⁴ The significant results of joint kinetics in the current study were above the measurement error for all variables, and above the MDC for KFM, KEM2, and KIRM. While this work contributes important new information towards understanding longer-term longitudinal changes in knee joint biomechanics after ACLR, future work incorporating imaging to determine knee joint degeneration is needed to understand if these biomechanical gait features are associated with the development of premature knee OA.

In conclusion, this work demonstrates longitudinal changes in both knee joint kinetics and kinematics from 2 to 8 years after ACLR surgery. The fact that similar changes were found on the contralateral knee, together with the lower rate of OA in the contralateral limb, suggests that the biomechanical changes are likely acting together with altered structural and biological factors in the ACLR knee. Given that approximately 50% of the ACLR population develop premature radiographic knee OA within 10–15 years of injury², the results of this study supports future larger and comprehensive investigations into long-term changes in joint mechanics in the ACLR population, including investigation of the clinical effect of these gait changes. Such work will help to identify specific features that place some patients at greatest risk of developing knee OA after ACLR.

Acknowledgments

The authors thank Sean Scanlan, PhD and Michael Zabala, PhD, for their assistance in data collection at baseline. This study was funded by the National Institute of Health grants AR039421 and AR052784. The funding sources had no role in the study design, collection or analysis of data, or manuscript preparation or submission.

References

1. Ardern CL, Webster KE, Taylor NF, et al. Return to sport following anterior cruciate ligament reconstruction surgery: a systematic review and meta-analysis of the state of play. *British journal of sports medicine*. 2011; 45:596–606. [PubMed: 21398310]
2. Lohmander LS, Ostengren A, Englund M, et al. High prevalence of knee osteoarthritis, pain, and functional limitations in female soccer players twelve years after anterior cruciate ligament injury. *Arthritis and rheumatism*. 2004; 50:3145–3152. [PubMed: 15476248]
3. Li RT, Lorenz S, Xu Y, et al. Predictors of radiographic knee osteoarthritis after anterior cruciate ligament reconstruction. *The American journal of sports medicine*. 2011; 39:2595–2603. [PubMed: 22021585]

4. Kessler MA, Behrend H, Henz S, et al. Function, osteoarthritis and activity after ACL-rupture: 11 years follow-up results of conservative versus reconstructive treatment. *Knee surgery, sports traumatology, arthroscopy : official journal of the ESSKA*. 2008; 16:442–448.
5. Weninger P, Zifko B, Liska M, et al. Anterior cruciate ligament reconstruction using autografts and double biodegradable femoral cross-pin fixation: functional, radiographic and MRI outcome after 2-year minimum follow-up. *Knee surgery, sports traumatology, arthroscopy : official journal of the ESSKA*. 2008; 16:988–995.
6. Kaur M, Ribeiro DC, Theis JC, et al. Movement Patterns of the Knee During Gait Following ACL Reconstruction: A Systematic Review and Meta-Analysis. *Sports medicine*. 2016; 46:1869–1895. [PubMed: 26936269]
7. Hart HF, Culvenor AG, Collins NJ, et al. Knee kinematics and joint moments during gait following anterior cruciate ligament reconstruction: a systematic review and meta-analysis. *British journal of sports medicine*. 2016; 50:597–612. [PubMed: 26265562]
8. Bullough PG, Yawitz PS, Tafta L, et al. Topographical variations in the morphology and biochemistry of adult canine tibial plateau articular cartilage. *Journal of orthopaedic research : official publication of the Orthopaedic Research Society*. 1985; 3:1–16. [PubMed: 3981289]
9. Zaid M, Lansdown D, Su F, et al. Abnormal tibial position is correlated to early degenerative changes one year following ACL reconstruction. *Journal of orthopaedic research : official publication of the Orthopaedic Research Society*. 2015; 33:1079–1086. [PubMed: 25721417]
10. Wellsandt E, Gardinier ES, Manal K, et al. Decreased Knee Joint Loading Associated With Early Knee Osteoarthritis After Anterior Cruciate Ligament Injury. *The American journal of sports medicine*. 2016; 44:143–151. [PubMed: 26493337]
11. Erhart-Hledik JC, Chu CR, Asay JL, et al. Gait mechanics 2 years after anterior cruciate ligament reconstruction are associated with longer-term changes in patient-reported outcomes. *Journal of orthopaedic research : official publication of the Orthopaedic Research Society*. 2017; 35:634–640. [PubMed: 27238273]
12. Andriacchi TP, Mundermann A. The role of ambulatory mechanics in the initiation and progression of knee osteoarthritis. *Current opinion in rheumatology*. 2006; 18:514–518. [PubMed: 16896293]
13. Andriacchi TP, Koo S, Scanlan SF. Gait mechanics influence healthy cartilage morphology and osteoarthritis of the knee. *The Journal of bone and joint surgery American volume*. 2009; 91(Suppl 1):95–101.
14. Shelbourne KD, Gray T. Minimum 10-year results after anterior cruciate ligament reconstruction: how the loss of normal knee motion compounds other factors related to the development of osteoarthritis after surgery. *The American journal of sports medicine*. 2009; 37:471–480. [PubMed: 19059893]
15. Tashman S, Kolowich P, Collon D, et al. Dynamic function of the ACL-reconstructed knee during running. *Clinical orthopaedics and related research*. 2007; 454:66–73. [PubMed: 17091011]
16. Papannagari R, Gill TJ, Defrante LE, et al. In vivo kinematics of the knee after anterior cruciate ligament reconstruction: a clinical and functional evaluation. *The American journal of sports medicine*. 2006; 34:2006–2012. [PubMed: 16885514]
17. Zabala ME, Favre J, Scanlan SF, et al. Three-dimensional knee moments of ACL reconstructed and control subjects during gait, stair ascent, and stair descent. *Journal of biomechanics*. 2013; 46:515–520. [PubMed: 23141637]
18. Butler RJ, Minick KI, Ferber R, et al. Gait mechanics after ACL reconstruction: implications for the early onset of knee osteoarthritis. *British journal of sports medicine*. 2009; 43:366–370. [PubMed: 19042923]
19. Varma RK, Duffell LD, Nathwani D, et al. Knee moments of anterior cruciate ligament reconstructed and control participants during normal and inclined walking. *BMJ open*. 2014; 4:e004753.
20. Webster KE, Feller JA. The knee adduction moment in hamstring and patellar tendon anterior cruciate ligament reconstructed knees. *Knee surgery, sports traumatology, arthroscopy : official journal of the ESSKA*. 2012; 20:2214–2219.

21. Patterson MR, Delahunt E, Caulfield B. Peak knee adduction moment during gait in anterior cruciate ligament reconstructed females. *Clinical biomechanics*. 2014; 29:138–142. [PubMed: 24359628]
22. Titchenal MR, Chu CR, Erhart-Hledik JC, et al. Early Changes in Knee Center of Rotation During Walking After Anterior Cruciate Ligament Reconstruction Correlate With Later Changes in Patient-Reported Outcomes. *The American journal of sports medicine*. 2017; 45:915–921. [PubMed: 27881382]
23. Webster KE, Feller JA, Wittwer JE. Longitudinal changes in knee joint biomechanics during level walking following anterior cruciate ligament reconstruction surgery. *Gait & posture*. 2012; 36:167–171. [PubMed: 22469773]
24. Tagesson S, Oberg B, Kvist J. Static and dynamic tibial translation before, 5 weeks after, and 5 years after anterior cruciate ligament reconstruction. *Knee surgery, sports traumatology, arthroscopy : official journal of the ESSKA*. 2015; 23:3691–3697.
25. Lin L, Li JS, Kernkamp WA, et al. Postoperative time dependent tibiofemoral articular cartilage contact kinematics during step-up after ACL reconstruction. *Journal of biomechanics*. 2016; 49:3509–3515. [PubMed: 27720228]
26. Knoll Z, Kocsis L, Kiss RM. Gait patterns before and after anterior cruciate ligament reconstruction. *Knee surgery, sports traumatology, arthroscopy : official journal of the ESSKA*. 2004; 12:7–14.
27. DeVita P, Hortobagyi T, Barrier J. Gait biomechanics are not normal after anterior cruciate ligament reconstruction and accelerated rehabilitation. *Medicine and science in sports and exercise*. 1998; 30:1481–1488. [PubMed: 9789847]
28. Wojtys EM, Huston LJ. Longitudinal effects of anterior cruciate ligament injury and patellar tendon autograft reconstruction on neuromuscular performance. *The American journal of sports medicine*. 2000; 28:336–344. [PubMed: 10843124]
29. Urbach D, Nebelung W, Weiler HT, et al. Bilateral deficit of voluntary quadriceps muscle activation after unilateral ACL tear. *Medicine and science in sports and exercise*. 1999; 31:1691–1696. [PubMed: 10613416]
30. Negahban H, Mazaheri M, Kingma I, et al. A systematic review of postural control during single-leg stance in patients with untreated anterior cruciate ligament injury. *Knee surgery, sports traumatology, arthroscopy : official journal of the ESSKA*. 2014; 22:1491–1504.
31. Lepley AS, Gribble PA, Thomas AC, et al. Quadriceps neural alterations in anterior cruciate ligament reconstructed patients: A 6-month longitudinal investigation. *Scandinavian journal of medicine & science in sports*. 2015; 25:828–839. [PubMed: 25693627]
32. McLeod S. *Sampling Methods*. 2014
33. Terrier P, Reynard F. Effect of age on the variability and stability of gait: a cross-sectional treadmill study in healthy individuals between 20 and 69 years of age. *Gait & posture*. 2015; 41:170–174. [PubMed: 25455699]
34. Andriacchi TP, Alexander EJ, Toney MK, et al. A point cluster method for in vivo motion analysis: applied to a study of knee kinematics. *Journal of biomechanical engineering*. 1998; 120:743–749. [PubMed: 10412458]
35. Favre J, Erhart-Hledik JC, Andriacchi TP. Age-related differences in sagittal-plane knee function at heel-strike of walking are increased in osteoarthritic patients. *Osteoarthritis and cartilage*. 2014; 22:464–471. [PubMed: 24445065]
36. Dyrby CO, Andriacchi TP. Secondary motions of the knee during weight bearing and non-weight bearing activities. *Journal of orthopaedic research : official publication of the Orthopaedic Research Society*. 2004; 22:794–800. [PubMed: 15183436]
37. Grood ES, Suntay WJ. A joint coordinate system for the clinical description of three-dimensional motions: application to the knee. *Journal of biomechanical engineering*. 1983; 105:136–144. [PubMed: 6865355]
38. Andriacchi, TPJT., Hurwitz, DE., Nataraja, RN. Musculoskeletal dynamics, locomotion, and clinical applications. In: Mow, VCHR., editor. *Basic Orthopaedic Biomechanics and Mechano-Biology*. 3. Philadelphia, PA: Lippincott Williams & Wilkins; 2005. p. 91-122.

39. Winter, DA. Biomechanics and motor control of human movement. 2. New York: Wiley-Interscience Publications; 1990.
40. Shi DL, Wang YB, Ai ZS. Effect of anterior cruciate ligament reconstruction on biomechanical features of knee in level walking: a meta-analysis. Chinese medical journal. 2010; 123:3137–3142. [PubMed: 21162970]
41. Scanlan SF, Chaudhari AM, Dyrby CO, et al. Differences in tibial rotation during walking in ACL reconstructed and healthy contralateral knees. Journal of biomechanics. 2010; 43:1817–1822. [PubMed: 20181339]
42. Zampeli F, Pappas E, Giotis D, et al. Kinematic predictors of subjective outcome after anterior cruciate ligament reconstruction: an in vivo motion analysis study. Knee surgery, sports traumatology, arthroscopy : official journal of the ESSKA. 2012; 20:785–792.
43. Kalawadia J, Thorhauer E, Arilla FV, et al. Knee Kinematics Are Related to Patient-Reported Outcomes 6 Months After Anatomic ACL Reconstruction. Orthopaedic Journal of Sports Medicine. 2015; 3:2325967115. S2325900031.
44. Henriksen M, Hinman RS, Creaby MW, et al. Rotational knee load predicts cartilage loss over 12 months in knee osteoarthritis. Osteoarthritis and Cartilage. 2012; 20:S17–S18.
45. Favre J, Erhart-Hledik JC, Chehab EF, et al. Baseline ambulatory knee kinematics are associated with changes in cartilage thickness in osteoarthritic patients over 5 years. Journal of biomechanics. 2016; 49:1859–1864. [PubMed: 27178021]
46. Chehab EF, Favre J, Erhart-Hledik JC, et al. Baseline knee adduction and flexion moments during walking are both associated with 5 year cartilage changes in patients with medial knee osteoarthritis. Osteoarthritis and cartilage. 2014; 22:1833–1839. [PubMed: 25211281]
47. Chang AH, Moio KC, Chmiel JS, et al. External knee adduction and flexion moments during gait and medial tibiofemoral disease progression in knee osteoarthritis. Osteoarthritis and cartilage. 2015; 23:1099–1106. [PubMed: 25677110]
48. Lewek M, Rudolph K, Axe M, et al. The effect of insufficient quadriceps strength on gait after anterior cruciate ligament reconstruction. Clinical biomechanics. 2002; 17:56–63. [PubMed: 11779647]
49. Andriacchi TP, Mundermann A, Smith RL, et al. A framework for the in vivo pathomechanics of osteoarthritis at the knee. Annals of biomedical engineering. 2004; 32:447–457. [PubMed: 15095819]
50. Henriksen M, Graven-Nielsen T, Aaboe J, et al. Gait changes in patients with knee osteoarthritis are replicated by experimental knee pain. Arthritis care & research. 2010; 62:501–509. [PubMed: 20391505]
51. Barenus B, Ponzer S, Shalabi A, et al. Increased risk of osteoarthritis after anterior cruciate ligament reconstruction: a 14-year follow-up study of a randomized controlled trial. The American journal of sports medicine. 2014; 42:1049–1057. [PubMed: 24644301]
52. Harkey MS, Luc BA, Golightly YM, et al. Osteoarthritis-related biomarkers following anterior cruciate ligament injury and reconstruction: a systematic review. Osteoarthritis and cartilage. 2015; 23:1–12. [PubMed: 25219671]
53. Chu CR, Andriacchi TP. Dance between biology, mechanics, and structure: A systems-based approach to developing osteoarthritis prevention strategies. Journal of orthopaedic research : official publication of the Orthopaedic Research Society. 2015; 33:939–947. [PubMed: 25639920]
54. Wilken JM, Rodriguez KM, Brawner M, et al. Reliability and Minimal Detectable Change values for gait kinematics and kinetics in healthy adults. Gait & posture. 2012; 35:301–307. [PubMed: 22041096]

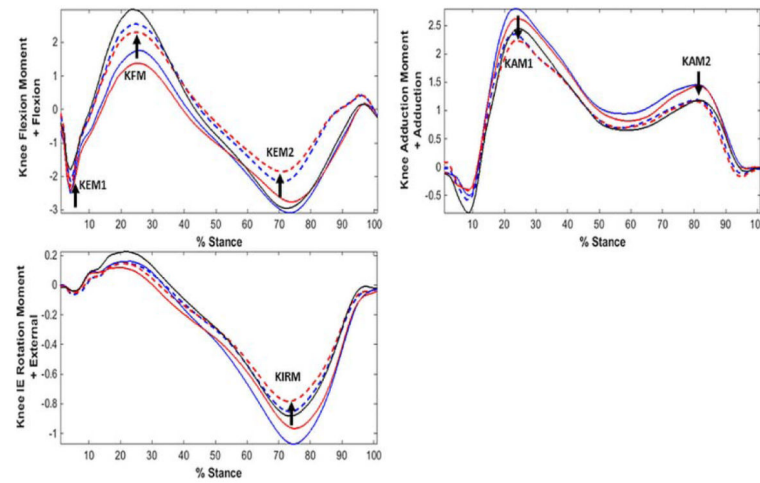


Figure 1.

The arrows indicate significant changes (Table 2) in the average flexion, adduction, and rotation moment curves (%Bw*Ht) of the ACLR (red) and contralateral (blue) limbs between 2 years (solid) and 8 years (dashed) post ACLR. The healthy control subjects are shown as a solid black line.

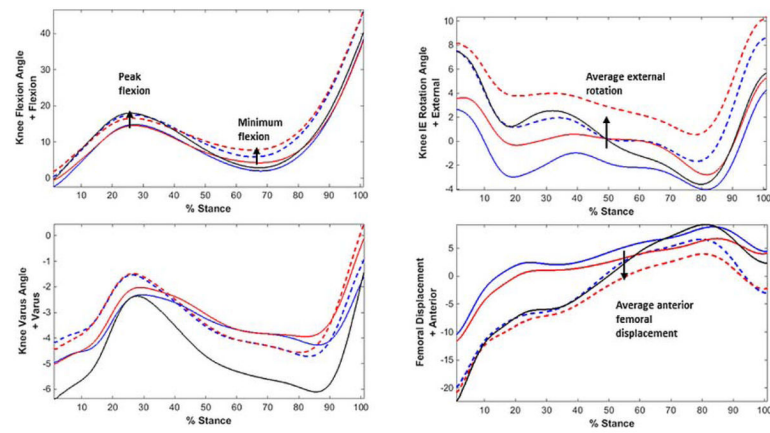


Figure 2.

The arrows indicate significant changes (Table 3) in average knee flexion, varus rotation, and internal rotation angles ($^{\circ}$), and anterior femoral displacement (mm) of the ACLR (red) and contralateral (blue) limbs between 2 years (solid) and 8 years (dashed) post ACLR. The healthy control subjects are shown as a solid black line.

Table 1

Demographics of the ACLR study participants at baseline (2.2±0.3 years post-ACLR) and follow-up (7.7±0.7 years post-ACLR). KOOS, IKDC, and Lysholm scores have a maximum value of 100, indicating best outcomes. Tegner scores range from 0 and 10, where 0 is ‘on sick leave/disability’ and 10 is ‘participation in competitive sports such as soccer at a national elite level’.

	Baseline	Follow-up
Age (yrs)	29.6 ± 7.3	35.1 ± 7.3
Weight (kg)	68.3 ± 10.2	72.4 ± 8.5
Height (m)	1.7 ± 0.1	1.7 ± 0.1
BMI (kg/m ²)	23.5 ± 2.7	25.0 ± 2.2
Lysholm score	94.0 ± 5.6	92.6 ± 9.2
Tegner score	---	5.3 ± 1.2
IKDC score	92.3 ± 8.2	91.1 ± 9.5
KOOS Pain score	95.5 ± 6.5	94.8 ± 5.8
KOOS Symptoms score	89.7 ± 13.9	88.8 ± 11.1
KOOS ADL score	99.5 ± 1.27	98.6 ± 3.2
KOOS Sports/Rec score	88.8 ± 18.4	92.2 ± 10.3
KOOS QOL score	79.8 ± 21.7	81.3 ± 16.9
KT-1000 side-to-side difference 30 lb manual force (mm)	0.6 ± 1.8	1.3 ± 2.3

All data given as mean ± standard deviation.

IKDC: International Knee Documentation Committee

KOOS: Knee Injury and Osteoarthritis Outcome Score

ADL: Activities of daily living; QOL: Quality of life

Tegner activity scale was not collected at baseline.

Table 2

Means (standard deviation) of knee joint moments in the ACLR and contralateral limbs in the ACL cohort at 2 and 8 years post-ACLR and in the healthy control knees. P-values shown are for results of repeated measures ANOVA and are adjusted for multiple comparisons with Holm-Bonferroni method. Bold indicates significance.

	2 years post-ACLR			8 years post-ACLR			Healthy Control		ANOVA Main Effect P-value		ANOVA Interaction P-value	
	ACLR	Contralateral		ACLR	Contralateral		Control		Time	Limb	Time x Limb	
KEM1	-3.09 (0.56)	-3.01 (0.59)		-2.72 (0.53)	-2.73 (0.42)		-2.46 (0.51)		0.048	0.787	1.000	
KFM	1.51 (1.22)	1.88 (1.16)		2.38 (1.01)	2.65 (0.90)		3.10 (0.51)		0.002	1.000	1.000	
KEM2	-2.81 (0.95)	-3.15 (0.83)		-1.96 (0.91)	-2.25 (0.81)		-3.06 (0.93)		<0.001	0.252	1.000	
KAM1	2.76 (0.37)	2.96 (0.39)		2.31 (0.39)	2.46 (0.52)		2.59 (1.02)		<0.001	0.208	1.000	
KAM2	1.54 (0.70)	1.59 (0.55)		1.28 (0.54)	1.30 (0.55)		1.33 (0.64)		0.009	1.000	1.000	
KIRM	-0.98 (0.22)	-1.09 (0.23)		-0.80 (0.22)	-0.87 (0.24)		-0.91 (0.24)		<0.001	0.208	1.000	
KERM	0.17 (0.09)	0.20 (0.10)		0.19 (0.08)	0.18 (0.08)		0.26 (0.08)		0.821	1.000	1.000	
KAM Impulse	0.76 (0.28)	0.79 (0.23)		0.60 (0.22)	0.63 (0.26)		0.63 (0.30)		0.004	1.000	0.856	

Units for joint moments are %Bw *Ht and for KAM Impulse are %Bw*Ht*s

Table 3

Means (standard deviation) of knee joint kinematics and spatiotemporal variables in the ACLR and contralateral limbs in the ACL cohort at 2 and 8 years post-ACLR and in the healthy control knees. P-values shown are for results of repeated measures ANOVA and are adjusted for multiple comparisons with Holm-Bonferroni method. Bold indicates significance.

	2 years post-ACLR				8 years post-ACLR		Healthy Control		ANOVA Main Effect P-value		ANOVA Interaction P-value	
	ACLR		Contralateral		ACLR	Contralateral	Control		Time	Limb	Time x Limb	
Peak flexion angle (°)	14.8 (5.4)	15.2 (6.6)	16.8 (4.6)	17.8 (3.9)	18.5 (6.1)				0.026	1.000	1.000	
Minimum flexion angle (°)	3.8 (5.3)	1.6 (4.1)	7.2 (4.1)	5.4 (3.8)	2.7 (4.2)				<0.001	0.045	1.000	
Peak varus angle (°)	-1.7 (2.1)	-2.0 (1.9)	-1.2 (3.1)	-1.3 (2.8)	1.8 (4.4)				0.680	0.773	1.000	
Peak valgus angle (°)	-4.4 (1.9)	-4.9 (2.4)	-4.9 (2.9)	-4.9 (2.8)	-6.5 (3.4)				0.530	1.000	1.000	
Average external rotation angle (°)	0.4 (3.3)	-1.5 (3.2)	3.8 (5.5)	1.7 (2.7)	1.3 (4.5)				0.007	0.160	1.000	
Average anterior femoral displacement (mm)	1.8 (4.6)	3.6 (5.9)	-3.6 (7.0)	-2.2 (4.3)	-1.3 (6.6)				0.006	0.98	0.889	
Speed (m/s)	1.4 (0.1)	1.3 (0.1)	1.3 (0.1)	1.3 (0.2)	1.4 (0.2)				1.000	0.354	0.496	
Step Length (m)	1.4 (0.1)	1.4 (0.1)	1.4 (0.1)	1.4 (0.1)	1.5 (0.1)				0.846	0.192	0.109	